

6.1 Project planning and scheduling

Project planning and scheduling are essential to the management of software development operations. Project planning is need to establishing the project's objectives, resources, tasks, hazards, and scope. It assures that it is clear what must be done and how. Project scheduling, on the other hand, arranges these tasks into a timeline that outlines when and how they should be finished. They both assist the development team in completing the project by the deadline.

6.2 Function Point Estimation

Function Point Analysis (FPA) is a standardized method for measuring the functional size of software applications. For the Hospital Management System, we will calculate the function points to estimate the project size, effort, and cost.

6.2.1. Functionality, Input, Output

The below table shows the functionality, input, and output of each function.

Table 6.1 Functionality, Input and Output.

Functionality	Input	Output
Patient Registration	Name, Email, Phone, Address, DOB, Gender, Blood Group, Emergency Contact	Patient account created, confirmation message or error
Doctor Registration	Name, Email, Phone, Specialization, Qualification, Experience, License Number, Department, Schedule	Doctor profile created, confirmation message
Staff Registration	Name, Email, Phone, Role, Department, Shift, Joining Date, Salary	Staff account created, confirmation message
User Authentication	Email, Password	Login successful, redirect to dashboard or error message
Appointment Booking	Patient ID, Doctor ID, Date, Time Slot, Request For, Symptoms, Payment Method	Appointment confirmed, receipt generated or error message
Doctor Schedule Management	Doctor ID, Day, Start Time, End Time, Max Patients, Status	Schedule updated, availability confirmation

Test Request Submission	Patient ID, Test Type, Description, Documents, Payment Method, Amount	Test request submitted, payment receipt or error
Medical Report Upload	Test Request ID, Report Title, Diagnosis, Notes, Report File	Report uploaded, patient notification sent
Payment Processing	User ID, Amount, Payment Method, Transaction ID, Gateway Response	Payment successful, receipt generated or transaction failed
Department Management	Department Name, Description, Icon, Head Doctor	Department created/updated, confirmation message
Leave Request	User ID, Leave Type, Start Date, End Date, Reason	Leave request submitted, pending approval notification
Appointment Cancellation	Appointment ID, Cancellation Reason	Appointment cancelled, refund processed or confirmation
Patient Search	Name, Phone, Patient Number	Patient list displayed with matching records
Doctor Search	Name, Specialization, Department	Doctor list with profiles and availability
View Appointment Details	Appointment ID	Complete appointment information, patient and doctor details
Check Doctor Availability	Doctor ID, Date	Available time slots, booking status
View Medical Report	Report ID	Report details, PDF download option, diagnosis and prescription
View Payment History	User ID	Transaction list, payment receipts, refund status
Staff Task Assignment	Staff ID, Task Title, Description, Deadline, Priority	Task assigned, notification sent to staff
Admin Dashboard	User activity logs, system statistics	Analytics, reports, user activity monitoring, system health
Email Notification	Recipient Email, Subject, Message, Appointment Details	Email sent, delivery confirmation

SMS Notification	Phone Number, Message, Appointment Reminder	SMS sent, delivery status
Generate Appointment Receipt	Appointment ID, Payment Details	PDF receipt with appointment and payment information
Generate Test Receipt	Test Request ID, Payment Details	PDF receipt with test details and payment confirmation
Generate Medical Report PDF	Report ID, Patient Details, Diagnosis	Formatted medical report PDF with hospital branding
User Profile Update	User ID, Updated Information	Profile updated, confirmation message
Password Change	Current Password, New Password, Confirm Password	Password updated successfully or validation error
View Staff Tasks	Staff ID	Assigned tasks list, deadlines, priorities, completion status
Department Statistics	Department ID	Patient count, doctor count, appointment statistics
User Activity Report	Date Range, User Role	Activity logs, login history, session duration, actions performed
Payment Reconciliation	Date Range, Payment Method	Transaction summary, total revenue, gateway-wise breakdown

6.2.2. Identify Complexity

Table 6.2 Identify Complexity of Data Function

Function Name	Types	Fields	DET	RET
Users	ILF	User ID, Name, Email, Password, Phone, Role, Email Verified, Phone Verified, Status, Created At, Updated At, Remember Token	12	1
Patients	ILF	Patient ID, User ID, Patient Number, DOB, Gender, Blood Group, Address, Emergency	13	1

		Contact, Emergency Phone, Medical History, Allergies, Current Medications, Insurance Info		
Doctors	ILF	Doctor ID, User ID, Doctor Number, Specialization, Qualification, Experience Years, License Number, Department ID, Consultation Fee, Visiting Days, Visiting Hours, Max Patients Per Day	12	1
Staff	ILF	Staff ID, User ID, Staff Number, Department ID, Position, Shift, Joining Date, Salary, Supervisor ID, Status	10	1
Departments	ILF	Department ID, Name, Description, Icon, Head Doctor ID, Status	6	1
Appointments	ILF	Appointment ID, Patient ID, Doctor ID, Appointment Date, Time Slot, Request For, Symptoms, Status, Payment Method, Payment Status, Amount, Notes	12	1
Test Requests	ILF	Test Request ID, Patient ID, Test Type, Description, Documents, Payment Method, Payment Status, Amount, Status, Admin Notes	10	1
Reports	ILF	Report ID, Test Request ID, Patient ID, Report Title, Diagnosis, Prescription, Doctor Comments, Admin Notes, Report File, Status	10	1
Payments	ILF	Payment ID, User ID, Payable Type, Payable ID, Amount, Payment Method, Transaction ID, Gateway Response, Status, Paid At	10	1
Doctor Schedules	ILF	Schedule ID, Doctor ID, Day of Week, Start Time, End Time, Max Patients, Is Available	7	1

Daily Tasks	ILF	Task ID, Staff ID, Assigned By, Task Title, Description, Priority, Status, Deadline	8	1
Leaves	ILF	Leave ID, User ID, Leave Type, Start Date, End Date, Total Days, Reason, Status, Approved By	9	1
Login Logs	ILF	Log ID, User ID, IP Address, User Agent, Login At, Logout At, Session Duration	7	1
Payment Gateway (bKash)	EIF	Transaction ID, Amount, Customer Phone, Status, Payment URL, Callback Data	6	1
Payment Gateway (Nagad)	EIF	Merchant ID, Order ID, Amount, Customer Phone, Payment Ref, Status	6	1
Payment Gateway (Rocket)	EIF	Transaction ID, Amount, Customer Account, Status, Reference	5	1
Email Service API	EIF	To, Subject, Body, Attachments, Status, Sent At	6	1
SMS Gateway API	EIF	Phone Number, Message, Status, Sent At, Delivery Status	5	1

Table 6.3 Identify Complexity of Transaction function

Function Name	Type	Fields	DET	FTR
Patient Registration	EI	Name, Email, Phone, Address, DOB, Gender, Blood Group, Emergency Contact, Password	9	2
Doctor Registration	EI	Name, Email, Phone, Specialization, Qualification, Experience, License Number, Department, Schedule, Password	10	3
Staff Registration	EI	Name, Email, Phone, Role, Department, Shift, Joining Date, Salary, Password	9	3

User Authentication	EI	Email, Password	2	2
Appointment Booking	EI	Patient ID, Doctor ID, Date, Time Slot, Request For, Symptoms, Payment Method, Amount	8	3
Doctor Schedule Update	EI	Doctor ID, Day, Start Time, End Time, Max Patients, Status	6	1
Test Request Submission	EI	Patient ID, Test Type, Description, Documents, Payment Method, Amount	6	2
Medical Report Upload	EI	Test Request ID, Report Title, Diagnosis, Notes, Report File, Status	6	1
Payment Processing	EI	User ID, Amount, Payment Method, Transaction ID, Gateway Response, Status	6	1
Department Creation	EI	Name, Description, Icon, Head Doctor, Status	5	1
Leave Request	EI	User ID, Leave Type, Start Date, End Date, Reason	5	2
Appointment Approval	EI	Appointment ID, Approved by Doctor, Approved by Admin	3	2
Leave Approval	EI	Leave ID, Approved by Admin	2	1
Appointment Cancellation	EI	Appointment ID, Cancellation Reason	2	1
Password Update	EI	Current Password, New Password, Confirm Password	3	1
Profile Update	EI	User ID, Name, Phone, Address, Other Details	5	2
Patient Dashboard	EO	Patient Info, Appointments, Reports, Test Requests, Payment History, Statistics	20	4
Doctor Dashboard	EO	Doctor Info, Schedule, Appointments, Patient List, Statistics	18	4

Admin Dashboard	EO	User Statistics, Appointment Analytics, Payment Summary, System Health, Activity Logs	25	10
Staff Dashboard	EO	Staff Info, Assigned Tasks, Schedules, Leave Status, Performance Metrics	15	4
Appointment Receipt PDF	EO	Appointment Details, Patient Info, Doctor Info, Payment Info, Hospital Branding	12	13
Test Request Receipt PDF	EO	Test Details, Patient Info, Payment Info, Hospital Branding	10	3
Medical Report PDF	EO	Report Details, Patient Info, Diagnosis, Prescription, Doctor Comments, Hospital Branding	15	4
User Activity Report	EO	User Info, Login Logs, Actions Performed, Session Duration, IP Address	12	3
Payment Reconciliation Report	EO	Transaction List, Total Revenue, Gateway-wise Breakdown, Date Range, Status Summary	14	3
Email Notification	EO	Recipient, Subject, Message, Appointment Details, Hospital Info	6	2
SMS Notification	EO	Phone Number, Message, Appointment Reminder	4	2
Search Patients	EQ	Search Query, Patient List with Name, Phone, Patient Number, DOB, Gender, Blood Group	7	1
Search Doctors	EQ	Search Query, Doctor List with Name, Specialization, Department, Qualification, Experience	6	2
View Appointment Details	EQ	Appointment ID, Complete Appointment Info, Patient Details, Doctor Details, Payment Status	12	3

Check Doctor Availability	EQ	Doctor ID, Date, Available Time Slots, Booking Status	4	3
View Medical Report	EQ	Report ID, Report Details, Diagnosis, Prescription, PDF Download Option	5	3
View Payment History	EQ	User ID, Transaction List, Payment Receipts, Refund Status	4	2
View Test Request Status	EQ	Test Request ID, Test Details, Status, Payment Info	4	2
View Staff Tasks	EQ	Staff ID, Task List, Deadlines, Priorities, Completion Status	5	2
View Department Details	EQ	Department ID, Department Info, Doctor List, Statistics	4	2
View User Profile	EQ	User ID, Profile Information	2	2

6.2.3 Unadjusted Function Point Contribution

Table 6.4 UFP of Transaction Function

Function Name	Type	DET	FTR	Complexity	UFP
Patient Registration	EI	9	2	Average	4
Doctor Registration	EI	10	3	High	6
Staff Registration	EI	9	3	High	6
User Authentication	EI	2	2	Low	3
Appointment Booking	EI	8	3	Average	4
Doctor Schedule Update	EI	6	1	Low	3
Test Request Submission	EI	6	2	Average	4
Medical Report Upload	EI	6	1	Low	3
Payment Processing	EI	6	1	Low	3
Department Creation	EI	5	1	Low	3
Leave Request	EI	5	2	Average	4
Appointment Approval	EI	3	2	Low	3

Leave Approval	EI	2	1	Low	3
Appointment Cancellation	EI	2	1	Low	3
Password Update	EI	3	1	Low	3
Profile Update	EI	5	2	Average	4
Patient Dashboard	EO	20	4	High	7
Doctor Dashboard	EO	18	4	High	7
Admin Dashboard	EO	25	10	High	7
Staff Dashboard	EO	15	4	High	7
Appointment Receipt PDF	EO	12	3	Average	5
Test Request Receipt PDF	EO	10	3	Average	5
Medical Report PDF	EO	15	4	High	7
User Activity Report	EO	12	3	Average	5
Payment Reconciliation Report	EO	14	3	Average	5
Email Notification	EO	6	2	Low	4
SMS Notification	EO	4	2	Low	4
Search Patients	EQ	7	1	Low	3
Search Doctors	EQ	6	2	Low	3
View Appointment Details	EQ	12	3	Average	4
Check Doctor Availability	EQ	4	3	Average	4
View Medical Report	EQ	5	3	Average	4
View Payment History	EQ	4	2	Low	3
View Test Request Status	EQ	4	2	Low	3
View Staff Tasks	EQ	5	2	Low	3
View Department Details	EQ	4	2	Low	3
View User Profile	EQ	2	2	Low	3

Total UFP (Transaction Function)	155
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Table 6.5 UFP of Data Function

Function Name	Type	DET	RET	Complexity	UFP
Users	ILF	12	1	Low	7
Patients	ILF	13	1	Low	7
Doctors	ILF	12	1	Low	7
Staff	ILF	10	1	Low	7
Departments	ILF	6	1	Low	7
Appointments	ILF	12	1	Low	7
Test Requests	ILF	10	1	Low	7
Reports	ILF	10	1	Low	7
Payments	ILF	10	1	Low	7
Doctor Schedules	ILF	7	1	Low	7
Daily Tasks	ILF	8	1	Low	7
Leaves	ILF	9	1	Low	7
Login Logs	ILF	7	1	Low	7
Payment Gateway (bKash)	EIF	6	1	Low	5
Payment Gateway (Nagad)	EIF	6	1	Low	5
Payment Gateway (Rocket)	EIF	5	1	Low	5
Email Service API	EIF	6	1	Low	5
SMS Gateway API	EIF	5	1	Low	5

Total UFP (Data Function)	116
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Table 6.6 Total Unadjusted Function Point (UFP)

Function Type	Count	Total UFP
External Input (EI)	16	59
External Output (EO)	11	63
External Inquiry (EQ)	10	33
Internal Logical File (ILF)	13	91
External Interface File (EIF)	5	25
Total Unadjusted Function Points (UFP)	55	271

6.2.4 Calculation of Total Degree of Influence (TDI)**Table 6.7 Calculation of Total Degree of Influence (TDI)**

SI	General System Characteristics	Brief Description	DI
01	Data Communication	System requires online communication between modules, payment gateways (bKash, Nagad, Rocket, DBBL, Mastercard, Visa), email/SMS services, and cloud storage.	5
02	Distributed Data Processing	Data is processed across vendors with cloud hosting, load balancing, and database replication.	3
03	Performance	The system must handle concurrent users efficiently with fast response times (target < 3 seconds), which is critical for hospital operations.	4
04	Heavily Used Configuration	Multiple user roles (patients, doctors, staff, admin) access the system simultaneously with high availability requirements.	4
05	Transaction Rate	High transaction rate due to appointment bookings, payments, staff operations, and concurrent users.	4
06	Online Data Entry	All data entry is performed online through web-based forms with no offline support.	5

07	End-User Efficiency	User-friendly interface with intuitive navigation, mobile responsiveness, and streamlined workflows for medical staff.	5
08	Online Update	Real-time updates for appointments, schedules, payments, inventory, and notifications across all user roles.	5
09	Complex Processing	Includes verification, payment gateway integration, PDF generation, and appointment scheduling algorithms.	4
10	Reusability	Core modules such as authentication, payments, and notifications are reusable across features.	3
11	Installation Ease	The system is designed for easy deployment and configuration on cloud or local servers.	3
12	Operational Ease	Administrative dashboard, automated backups, and monitoring tools simplify daily operations.	4
13	Multiple Sites	Designed for single-hospital deployment with potential for multi-branch expansion.	2
14	Facilitate Change	Modular design supports future upgrades, new features, and vendor expansion.	4

Total TDI	55
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6.2.5. Final Calculation

$$\text{Value Adjustment Factor (VAF)} = 0.65 + (0.01 \times \text{TDI})$$

$$= 0.65 + (0.01 \times 55)$$

$$= 1.10$$

$$\text{Total UFP} = \text{UFP (TF)} + \text{UFP (DF)}$$

$$= 155 + 116$$

$$= 271$$

$$\begin{aligned}
 \text{AFP (Adjusted Function Point)} &= \text{UFP (Total)} * \text{VAF} \\
 &= 271 * 1.10 \\
 &= 298.1
 \end{aligned}$$

6.3 Project Scheduling

We have to complete this project within 12 weeks or 3 Month. So, we have created a Gantt Chart for our use in time management. Our whole timeline is shown in below Gantt Chart.

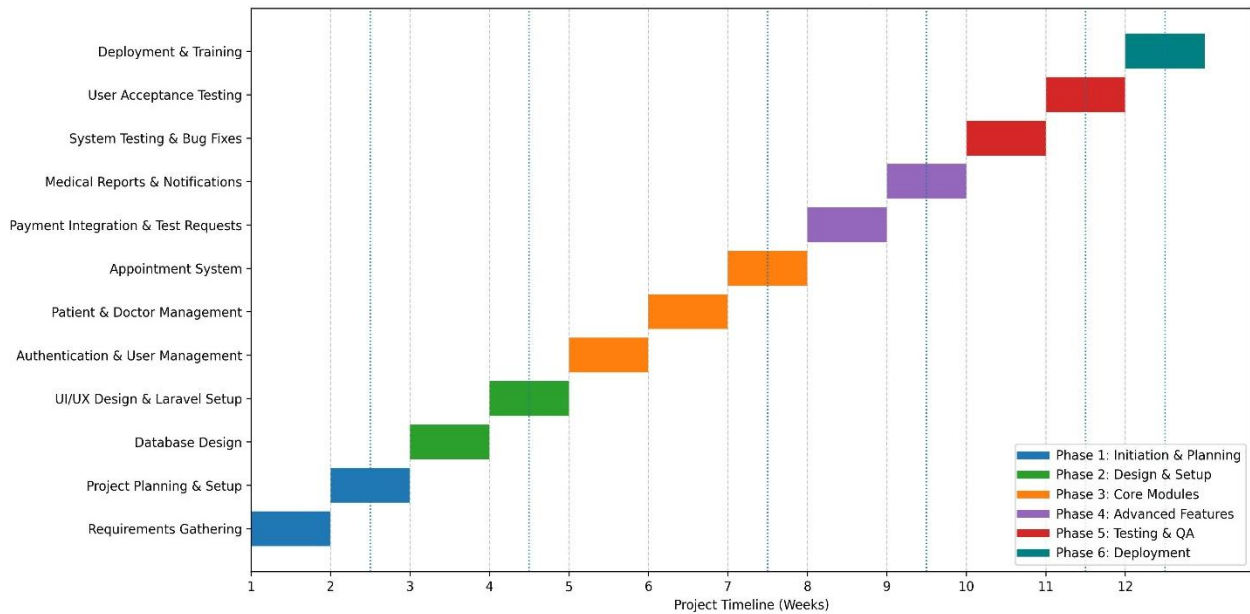


Figure 6.1 Gantt Chart

This Figure 6.1 shows the hospital management system project is divided into six separate phases and has a planned 12-week development timetable. Weeks 1-3 of the project requirement gathering and Laravel setup. Weeks 3-5 are database design UI/UX design and Laravel setup. The appointment system, patient & doctor management are all included in the core development phase (Weeks 5-8). Weeks 8-10 saw the implementation of advance features like payment integration & test request and medical report & notification. System testing and bug fixes, user acceptance testing and deployment and training are the main topics of the last phase (Weeks 10-12). Throughout the project lifecycle are iterative testing and quality